

- 6 (a) Collagen is the most common structural protein in vertebrates. Collagen provides the skin with flexibility and strength.

Explain how the structure of a collagen fibre provides the skin with strength.

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- (b) The enzyme collagenase breaks down collagen. Collagenase has several important medical uses, such as in the treatment of burnt skin.

Scientists investigated the effect of pH on the activity of collagenase at 37 °C.

The results of their investigation are shown in Fig. 6.1.

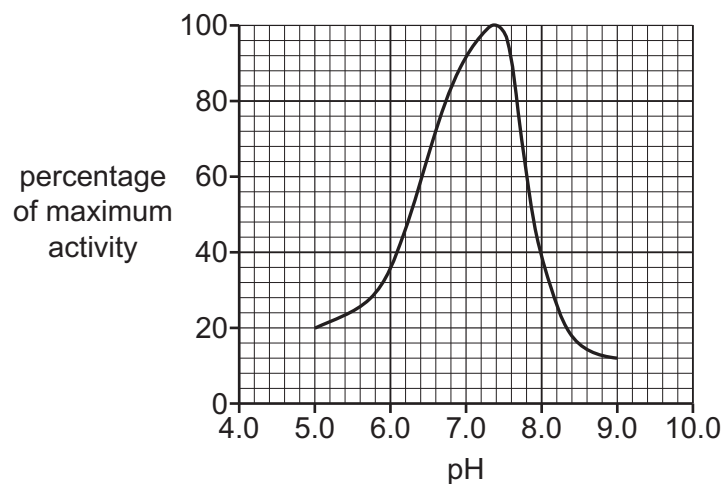


Fig. 6.1

Explain why the activity of collagenase is lower at pH 8.0 than at the optimum pH.

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[Total: 5]